Arun Pal

Senior Java Backend Engineer | Microservices · Cloud · AI

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8+ Years Experience

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PROFESSIONAL SUMMARY

Senior Java Backend Engineer with 8+ years of experience delivering scalable Java microservices and production AI systems. Deep expertise in Java, Spring Boot, Micronaut, REST APIs, multi-threading, JVM tuning, and AWS/Kubernetes deployment – driving 30% faster API response times, 25% fewer integration defects, and 40% fewer production incidents. Built a production RAG chatbot on AWS Bedrock (100+ concurrent users, 99.5% uptime) and Python MCP servers that let AI agents automate infrastructure.

TECHNICAL SKILLS

Java & Backend: Java (primary, 8+ yrs), Spring Boot, Spring Cloud, Micronaut, Hibernate, REST APIs, Microservices, Multi-threading, OAuth2, JWT

Other Languages & UI: Python, Go, JavaScript, SQL, Angular, ExtJS, jQuery

Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, CI/CD, GitLab, GitHub Actions, SonarQube, Git, Datadog, Power Automate, CloudWatch, Linear, Jira

Data & Messaging: MySQL, PostgreSQL, Oracle DB, MongoDB, Redis, Apache Kafka, Hadoop, Apache Spark, Airflow, Temporal, Databricks

Testing, Resilience & Observability: JUnit 5, TestNG, Mockito, Gatling, Resilience4j, OpenTelemetry, Prometheus, JVM tuning, profiling, system design, thread-dump analysis, Blackduck, Checkmarx

AI & LLM: AWS Bedrock, RAG, FAISS, Titan Embeddings, Prompt Engineering, GitHub Copilot, Claude AI (Spec-Driven & Conversational Development Workflows), LLM-Assisted Code Generation, Python MCP Servers

PROFESSIONAL EXPERIENCE

Precisely Software & Data Solutions, Noida, India — *Oct 2021 – Present*

Senior Software Engineer

- Architected Java-based RESTful microservices with OAuth2/JWT security, robust error handling, and seamless microservices integration, reducing integration defects by 25%.
- Engineered high-performance Java APIs using multi-threading, Redis caching, and optimized DB query patterns, improving average API response time by 30%.
- Owned deployment of cloud-native Java services on AWS and Kubernetes, designing API Gateway rate limiting, DDoS protection, and centralized logging to harden the security posture.
- Led code reviews across multiple modules, enforcing SOLID principles, clean-code, and performance standards that cut production incidents by 40%.
- Mentored junior engineers through structured code reviews, pair programming, and Agile best practices, improving delivery quality and onboarding speed.
- Built end-to-end observability with Datadog dashboards and alerting; integrated JUnit 5 and Gatling into CI/CD pipelines, improving deployment reliability and incident response.
- Adopted spec-driven, LLM-assisted development with GitHub Copilot and Claude AI to accelerate code generation and boilerplate delivery while enforcing quality through peer review.
- Drove engineering workflow automation via Linear and GitHub Actions integration (PR-to-issue linking, status sync) and automated MS Teams pipeline notifications.

Trangile Software Solution, Noida, India — *Feb 2020 – Sep 2021*

Software Engineer

- Collaborated with cross-functional stakeholders to gather requirements and produce Technical Specification Documents (TSDs) aligned with project goals.
- Developed Java and Oracle-based backend modules per technical documentation, ensuring adherence to quality and coding standards.
- Integrated Advance Deposit and Prepaid Card payment modules, enabling seamless payment workflows across multiple applications.
- Led the migration of applications from OC4J to JBoss and upgraded Java 5 to Java 8, maintaining Oracle DB compatibility with zero production downtime.
- Authored code review reports and release documentation for UAT and production deployments, improving release traceability.

Vinculum Software Solution, Noida, India
Associate Software Engineer
Jul 2018 – Jan 2020

- Assisted senior engineers in designing and developing Java and Oracle-based backend solutions for e-commerce logistics platforms.
- Contributed to technical documentation, deployment runbooks, and release processes, gaining hands-on experience with the full SDLC, Agile ceremonies, and enterprise Java practices.

KEY PROJECTS

Backend & Data Engineering *Java · AWS S3 · Kafka · Apache Spark · Temporal · Python*

- **Spring Cloud API Gateway:** Built a production-ready gateway as the single microservices entry point with dynamic YAML routing (hot-reload), multi-tenant JWT validation, Redis-backed IP- and user-based rate limiting, Resilience4j circuit breaking, and OpenTelemetry tracing with Prometheus metrics.
- **Bulk File Processing (Kafka · Spark · Temporal):** Architected an asynchronous large-file processing platform on AWS S3, Apache Kafka, Apache Spark, and Temporal Workflow, delivering chunk-based processing, multipart uploads, fault-tolerant workflow retries, and secure pre-signed URL CSV exports.
- **DIS Data Quality Pipeline (Spark Streaming · Databricks):** Built Spark Streaming and Databricks pipelines with grouping and geocoding operators to automate data-quality improvements and significantly reduce manual correction effort.
- **Swagger Validation Framework:** Built an automated framework to validate Swagger schemas, examples, and generated clients, ensuring production-ready API specifications and consistent API responses.
- **Runtime Data Pipeline:** Implemented an Excel-driven CI/CD pipeline that uploads datasets to S3 and triggers automated data refresh workflows for the Addressing API without code changes.
- **S3 CSV Chunk Processing:** Engineered a high-performance CSV reader for large S3 files, processing data in non-overlapping chunks with full multi-byte character support and zero data loss.

AI-Powered Documentation Assistant *AWS Bedrock · RAG · LLM · Production*

- **RAG Pipeline:** Built semantic search using FAISS, document chunking, and AWS Bedrock Titan embeddings.
- **Scale & Accuracy:** Supported 100+ concurrent users at 99.5% uptime and 95% answer accuracy using async/await, semaphore-based concurrency control, and WebSocket streaming.
- **Cost Optimization:** Cut AI infrastructure costs by 60% through Redis caching, input deduplication, and tiered model selection.
- **Reliability:** Implemented LLM-based confidence scoring, hallucination detection, and source citation, plus circuit breaker, exponential backoff, graceful degradation, and CloudWatch monitoring with distributed tracing.

MCP Tool Suite — Geocoding, API Gateway & Service Provisioning *Python · MCP · AWS · Agentic AI*

- **Geocoding MCP:** Exposed geocoding APIs as MCP tools so AI agents autonomously select the correct MCP, construct requests, and return structured address results.
- **API Gateway Route Manager MCP:** Lets agents add API Gateway routes and enable Ingress endpoints directly, making live infrastructure changes with no code checkout or manual deployment.
- **Service Provisioning MCP:** Automates multi-region service setup (e.g. US, AUS): creates the required repositories, configures data upload pipelines, and provisions region-specific geocoding services end-to-end.

AI Service Bootstrapper — Design Doc to Production-Ready Service *LLM · Agentic AI · GitLab · Jira · API Gateway*

- **Design-Doc Parsing:** Reads a SolDef technical design document and extracts the service name, tech stack, request/response schemas, and infrastructure needs (e.g. Redis, database).
- **Automated Delivery:** Creates the GitLab project and repository with generated APIs, database connections, and deployment files; exposes APIs through the API Gateway and raises PRs, cutting a 2–3 day setup to under an hour.
- **Ticketing & Reliability Docs:** Raises Jira tickets, including platform-team infrastructure tickets, and drafts SLI/SLO documents with PRs.

Power Automate Workflow Suite *Power Automate · MS Teams · CI/CD*

- **Leave Notifications:** Automated same-day and next-day leave alerts to the team channel for daily capacity visibility.
- **Sprint Overflow Analyzer:** Cross-references team leave schedules with public holidays to surface sprint capacity risk in Teams before sprint start.
- **CI/CD Pipeline Notifier:** Posts adapter run status, failure alerts, and PR summaries with direct links to Teams for one-click review and faster merges.

EDUCATION & CERTIFICATIONS

Dr. APJ Abdul Kalam Technical University
B.Tech, Computer Science & Engineering · CGPA 7.2 · Aug 2014 – Jul 2018

HackerRank: Java, Problem Solving, SQL, CSS
Udemy: Spring Boot Foundation, Micronaut, AWS